

COMP5310 26s1 Final Practice

USYD 各科往届题库



W1 Intro Ds And Big Data

Q1. Short Answer Semester 2 2023 Final (modified)

Give two reasons why design thinking is useful when formulating an authentic data science problem.

Q2. Short Answer Semester 2 2023 Final (modified)

Choose two common data science pitfalls, such as using bad data, confusing correlation with causation, torturing data, or doing harm. For each, give a concrete example of a bad outcome.

Q3. MCQ

A supermarket wants to predict which customers will respond to a coupon before sending it. Which framing is most appropriate?

- (A) Unsupervised clustering only
- (B) Supervised binary classification
- (C) Association rules with no target
- (D) Data warehousing only

Q4. Short Answer

List the typical data science workflow in order and give one quality risk at the data acquisition stage.

Q5. MCQ

Which variable type is “suburb” in a customer table?

- (A) Nominal
- (B) Ordinal
- (C) Interval
- (D) Ratio

Q6. Short Answer

Explain why a technically accurate model can still fail as a decision-support tool.

Q7. MCQ

Which statement best distinguishes data science from simple reporting?

- (A) It always uses neural networks
- (B) It connects data management, analysis, modelling and decisions
- (C) It avoids databases
- (D) It only studies causal experiments

Q8. Short Answer

A table contains one row per transaction, not per customer. Why does the row grain matter before modelling churn?

Q9. MCQ

Which big-data V mainly describes fast incoming sensor observations?

- (A) Volume
- (B) Velocity
- (C) Variety
- (D) Veracity

Q10. Short Answer

Give two examples of data leakage in a predictive task.



W2 Spreadsheet Eda

Q1. MCQ Semester 1 2024 Final (modified)

A numeric goodness-of-fit measure is reported for a linear regression model. Which option gives the usual ordinary least-squares regression with an intercept?

- (A) From 0 to 1
- (B) Always greater than 1
- (C) Always less than 0
- (D) It must equal the correlation coefficient

Q2. Calculation Semester 2 2023 Final (modified)

A survey samples residents from towns A, B and C with populations 10k, 30k and 40k. To construct a stratified subsample of size 160 proportional to town population, how many residents should be selected from each town?

Q3. MCQ

Which statistic is most suitable for the centre of a heavily right-skewed income distribution?

- (A) Mean
- (B) Median
- (C) Range
- (D) Standard deviation

Q4. Calculation

For values 2, 3, 3, 4, 20, compute the mean and median, then state which better summarises a typical value.

Q5. MCQ

Which plot is most appropriate for counts of nominal categories?

- (A) Bar chart
- (B) Scatter plot
- (C) Histogram
- (D) Line chart

Q6. Short Answer

Explain why nominal data should not be interpreted as naturally ordered after label encoding.

Q7. Calculation

A variable has mean 50 and standard deviation 10. Compute the z-score for $x = 65$.

Q8. MCQ

Pearson correlation is most appropriate for which relationship?

- (A) Linear relation between numeric variables
- (B) Any nominal variables
- (C) Any causal claim
- (D) Ranking without numeric distance

Q9. Short Answer

When would Spearman correlation be preferable to Pearson correlation?

Q10. MCQ

For a left-skewed numeric variable, which ordering is typical?

- (A) mean < median < mode
- (B) mode < median < mean
- (C) mean = median = mode
- (D) median < mean < mode



W3 Python Eda

Q1. Table/Diagram Semester 2 2023 Final (modified)

The table below lists common data-quality problems. For any two, describe one detection or measurement method.

Problem	Example check
Incomplete	missing-rate by column
Inconsistent	conflicting units/codes
Duplicate	repeated business key
Out-of-date	stale timestamp
Invalid	value outside allowed range

Q2. MCQ

Which Pandas operation is most directly used to summarise sales by month?

- (A) groupby
- (B) dropna
- (C) head
- (D) rename

Q3. Short Answer

Why should scaling parameters be fitted on the training set only?

Q4. Calculation

A dataset has 2000 rows and 140 missing values in column A. Compute the missing-rate percentage.

Q5. MCQ

Which missing-data mechanism means missingness is unrelated to observed or unobserved values?

- (A) MCAR
- (B) MAR
- (C) MNAR
- (D) PCA

Q6. Short Answer

Give one advantage and one risk of deleting all rows with missing values.

Q7. MCQ

Which encoding is generally appropriate for a nominal variable with levels red, blue, green?

- (A) One-hot encoding
- (B) Treat as 1,2,3 ordered scale
- (C) Do not encode
- (D) Use the target values directly

Q8. Short Answer

Explain what a fixed random seed helps with and what it does not guarantee.

Q9. MCQ

Which split should be used only once for the final unbiased estimate?

- (A) Training
- (B) Validation
- (C) Test
- (D) Bootstrap sample

Q10. Short Answer

Name two checks for duplicate records and why the duplicate definition depends on business rules.



W4 Clean And Store

Q1. Short Answer Semester 2 2024 Final (modified)

A warehouse designer can use either a star schema or a snowflake schema for enrolment analytics. Explain the differences and one practical trade-off.

Q2. MCQ

In a relational table, what does a foreign key enforce?

- (A) Referential integrity
- (B) A unique row order
- (C) Normal distribution
- (D) Higher accuracy

Q3. Short Answer

Explain the difference between a fact table and a dimension table in a star schema.

Q4. MCQ

Which schema usually uses denormalised dimension tables around a central fact table?

- (A) Star schema
- (B) Snowflake schema
- (C) Heap file
- (D) Decision tree

Q5. Short Answer

State two advantages and one disadvantage of a snowflake schema relative to a star schema.

Q6. Calculation

A sales fact table has 5000 rows and each row is one product line in a transaction. If a transaction has 3 products, how many fact rows represent it? Explain the grain.

Q7. MCQ

Which task belongs to the transform stage of ETL?

- (A) Standardising date formats
- (B) Buying a server only
- (C) Presenting final slides
- (D) Choosing a p-value

Q8. Short Answer

Why can a surrogate key be useful even when a natural key exists?

Q9. MCQ

Which problem occurs if a fact row references a non-existent dimension row?

- (A) Orphan foreign key
- (B) Multicollinearity
- (C) Type I error
- (D) High recall

Q10. Short Answer

Define the grain for a warehouse table storing lecture attendance.



W5 Sql Summary

Q1. Short Answer Semester 1 2023 Final (modified)

A Store table, Tx table, Tx_Item bridge table, and Item table record transactions and item names. Describe the join to list unique item names sold in March.

Q2. Calculation Semester 2 2022 Final (modified)

Tables Station(Station, SiteName, Commence, OrgCode) and Measurement(Station, Sensor, Date, Value) are given. Write an SQL query listing each SiteName with the total number of measurements and average measurement value.

Q3. MCQ

In SQL, which clause filters groups after aggregation?

- (A) WHERE
- (B) HAVING
- (C) FROM
- (D) LIMIT

Q4. Calculation

A joined result has three rows with values NULL, 10, and 50 in column A. What are COUNT(*) and COUNT(A)?

Q5. MCQ

A LEFT JOIN from a 10-row table to an empty right table returns what?

- (A) 0 rows
- (B) 10 rows with NULLs for right columns
- (C) 10 rows without NULLs
- (D) SQL error

Q6. Short Answer

Explain why NATURAL JOIN can be risky in production SQL.

Q7. Calculation

Write a query pattern that returns films with more than five actors, grouped by film id and title.

Q8. MCQ

Which SQL expression extracts the month in standard SQL-style syntax?

- (A) EXTRACT(MONTH FROM date_col)
- (B) MONTH = date_col
- (C) COUNT(MONTH)
- (D) JOIN MONTH

Q9. Short Answer

Differentiate WHERE and HAVING using one example.

Q10. MCQ

Which aggregate ignores NULL values in its argument?

- (A) COUNT(col)
- (B) COUNT(*)
- (C) SELECT *
- (D) LEFT JOIN



W6 Hyp Test And Eval

Q1. MCQ Semester 1 2024 Final (modified)

In a hypothesis test, suppose H_0 is actually true but the test rejects H_0 . What has occurred?

- (A) Type I error
- (B) Type II error
- (C) Correct non-rejection
- (D) Correct rejection

Q2. Short Answer Semester 1 2023 Final (modified)

A team can launch only one of two new drink flavours. Design an experiment to determine which flavour is likely to be received more favourably. State the research question, H_0 , response measurement, and test choice.

Q3. MCQ

If H_0 is true but a test rejects H_0 , what error occurred?

- (A) Type I error
- (B) Type II error
- (C) No error
- (D) Overfitting

Q4. Calculation

At significance level $\alpha = 0.01$, a test gives $p = 0.005$. State the decision and interpretation.

Q5. Short Answer

Write a null and alternative hypothesis for testing whether two independent groups have different mean shopping times.

Q6. MCQ

Which metric is most relevant when false positives are very costly?

- (A) Precision
- (B) Recall
- (C) Training loss only
- (D) Number of features

Q7. Calculation

A classifier has TP=30, FP=10, FN=20, TN=40. Compute accuracy, precision, and recall.

Q8. Short Answer

Explain why k-fold cross-validation helps detect but does not itself solve overfitting.

Q9. MCQ

Which test is commonly used for association between two categorical variables?

- (A) Chi-square test
- (B) Linear regression SSE
- (C) k-means
- (D) PCA

Q10. Short Answer

State two assumptions or design checks before choosing a two-sample t-test.



W7 Assoc And Pca

Q1. Calculation Semester 2 2022 Final (modified)

Transactions are: T1 Apple, Banana, Cheese, Eggs; T2 Apple, Banana, Cheese, Diapers, Eggs; T3 Apple, Banana, Milk; T4 Apple, Banana, Diapers. With $\text{min_sup}=60\%$, list all frequent 2-itemsets.

Q2. Short Answer Semester 2 2024 Final (modified)

PCA on 100 features keeps 95% variance and removes 10 principal components. What does this imply, and why should we not say that 10 original variables were simply deleted?

Q3. MCQ

For rule $X \rightarrow Y$, confidence is:

- (A) $P(Y|X)$
- (B) $P(X)$
- (C) $P(Y)$
- (D) $P(X)/P(Y)$

Q4. Calculation

In 100 transactions, A appears in 40, B appears in 50, and both appear in 30. Compute support and confidence for $A \rightarrow B$.

Q5. Short Answer

Explain the Apriori downward-closure property.

Q6. MCQ

Which method mines frequent patterns using a compressed prefix tree?

- (A) FP-growth
- (B) k-means
- (C) logistic regression
- (D) t-test

Q7. Short Answer

Why can high-confidence rules still be uninteresting? Mention lift or base rate.

Q8. MCQ

PCA is best described as:

- (A) Unsupervised dimensionality reduction
- (B) Supervised classification
- (C) SQL aggregation
- (D) A hypothesis test

Q9. Calculation

Eigenvalues of four PCs are 5, 3, 1, 1. How many PCs are needed to retain at least 80 percent variance?

Q10. Short Answer

Why should numeric features often be standardised before PCA?



W8 Clustering

Q1. Table/Diagram Semester 2 2024 Final (modified)

Four observations have pairwise distances $d(A, B) = 2$, $d(A, C) = 6$, $d(A, D) = 10$, $d(B, C) = 5$, $d(B, D) = 9$. Using single linkage, perform the first two merges and sketch the merge structure.



Q2. Short Answer Semester 2 2022 Final (modified)

Describe the procedure of a prototype-based clustering algorithm, and name one way to evaluate or choose the number of clusters.

Q3. MCQ

In k-means, the number of clusters k is:

- (A) An input parameter
- (B) Always learned automatically
- (C) A p-value
- (D) A class label

Q4. Calculation

For one-dimensional points 0, 2, 10, 12 with centroids 1 and 11, assign points to clusters and update centroids.

Q5. Short Answer

Give two reasons k-means can produce poor clusters.

Q6. MCQ

Which linkage uses the smallest pairwise distance between two clusters?

- (A) Single linkage
- (B) Complete linkage
- (C) Average linkage
- (D) Ward linkage

Q7. Calculation

If within-cluster SSE values for $k = 1, 2, 3, 4$ are 100, 45, 30, 27, which k might the elbow suggest? Explain.

Q8. Short Answer

Interpret a silhouette score close to 1 and one close to -1.

Q9. MCQ

Which preprocessing is especially important before distance-based clustering of variables with different units?

- (A) Standardisation
- (B) Adding IDs
- (C) Increasing p-value
- (D) Natural join

Q10. Short Answer

Why is clustering normally considered exploratory rather than confirmatory?



W9 Regression

Q1. Short Answer Semester 2 2024 Final (modified)

Compare simple linear regression and multiple linear regression in functional form and coefficient interpretation, when predictors in the multiple model are highly correlated.

Q2. True/False Semester 2 2024 Final (modified)

True or False: if adding one feature increases training R^2 , the new regression model must generalise better. Explain.

Q3. MCQ

Which value is guaranteed not to decrease on the training set when adding predictors to ordinary least squares?

- (A) R^2
- (B) Test RMSE
- (C) Test accuracy
- (D) p-value

Q4. Calculation

For observed $y = (2, 4, 5)$ and predictions $(1, 5, 5)$, compute SSE.

Q5. Short Answer

Explain multicollinearity and why a coefficient can be near zero even when the predictor has high pairwise correlation with y .

Q6. MCQ

Which regularisation tends to set some coefficients exactly to zero?

- (A) L1/lasso
- (B) L2/ridge
- (C) No penalty
- (D) k-means

Q7. Calculation

For $y = 1.115x - 1.077$, compute the prediction when $x = 4$ and the squared error if observed $y = 3.1$.

Q8. Short Answer

Differentiate simple linear regression and multiple linear regression in coefficient interpretation.

Q9. MCQ

Which residual pattern suggests nonlinearity?

- (A) Curved residual trend
- (B) Random cloud around zero
- (C) Constant variance
- (D) No residuals

Q10. Short Answer

Why can adjusted R^2 be more useful than R^2 for comparing models with different numbers of predictors?



W10 Classification

Q1. Calculation Semester 2 2022 Final (modified)

A decision-tree dataset predicts whether a person watches a football game using Weekend, Friends, and Star. information gain would be used to select the root split, and what must be computed for each attribute.

Q2. True/False Semester 1 2024 Final (modified)

True or False: gradient descent is guaranteed to find the global optimum for any differentiable loss function. Briefly justify.

Q3. MCQ

The logistic regression output before thresholding is best interpreted as:

- (A) Estimated class probability
- (B) Cluster id
- (C) SQL count
- (D) PCA eigenvalue

Q4. Calculation

Compute $\sigma(0) = 1/(1 + e^0)$ and classify with threshold 0.5.

Q5. Short Answer

Why must the logistic loss be differentiable for gradient-based optimisation?

Q6. MCQ

What is the main purpose of pruning a decision tree?

- (A) Reduce overfitting
- (B) Increase training accuracy only
- (C) Remove the target
- (D) Standardise variables

Q7. Calculation

A binary node has 6 positives and 2 negatives. Compute entropy to two decimals.

Q8. Short Answer

Explain information gain in a decision tree split.

Q9. MCQ

Naive Bayes assumes features are conditionally independent given the:

- (A) Class
- (B) Row number
- (C) SQL table
- (D) Cluster centroid

Q10. Calculation

A class prior is 0.6 and three conditional likelihoods are 0.5, 0.2, 0.8. Compute the unnormalised Naive Bayes score.



W11 Text And Bayes

Q1. Calculation Semester 2 2022 Final (modified)

Two documents are $D1 = \text{snorkel, around, reef}$, $D2 = \text{enter, snorkel, submarine}$. Query $Q = \text{snorkel, submarine}$. term vectors over snorkel, submarine, compute the Manhattan distance between Q and $D1$.

Q2. Calculation Semester 2 2024 Final (modified)

A Naïve Bayes classifier has $P(\text{yes}) = 0.5$, $P(\text{head} = \text{narrow}|\text{yes}) = 0.8$, $P(\text{size} = \text{large}|\text{yes}) = 0.4$, $P(\text{color} = \text{white}|\text{yes}) = 0.25$. Compute the unnormalised score for class yes for an animal with narrow head, large size, and white colour.

Q3. MCQ

In TF-IDF, a term appearing in nearly every document receives:

- (A) Low IDF weight
- (B) Infinite IDF weight
- (C) A class label
- (D) A p-value

Q4. Short Answer

Give two limitations of bag-of-words text representation.

Q5. Calculation

With $N = 1000$ documents and a term in $df = 10$ documents, compute $\log(N/df)$ using natural log approximately.

Q6. MCQ

Which model is often strong and simple for high-dimensional sparse text classification?

- (A) Naive Bayes
- (B) k-means only
- (C) Star schema
- (D) t-test only

Q7. Short Answer

Explain why removing header fields can prevent leakage in spam classification.

Q8. MCQ

Laplace smoothing in Naive Bayes mainly handles:

- (A) Unseen category probabilities
- (B) SQL joins
- (C) PCA variance
- (D) Dendrogram height

Q9. Calculation

For binary class spam, prior 0.4, $P(\text{free}|\text{spam}) = 0.5$, $P(\text{offer}|\text{spam}) = 0.3$. Compute the unnormalised score for a document containing both words.

Q10. Short Answer

Why might precision be more important than recall for an automatic spam deletion system?



W12 Ethics And Comms

Q1. MCQ

Which is an ethical risk of using historical hiring data to train a screening model?

- (A) Historical bias can be reproduced
- (B) The database becomes normalised
- (C) The model always becomes causal
- (D) SQL cannot aggregate

Q2. Short Answer

Explain data minimisation and why it matters for privacy.

Q3. MCQ

Which claim requires more than correlation from observational data?

- (A) Causal effect claim
- (B) Descriptive mean
- (C) Count of rows
- (D) Missing rate

Q4. Short Answer

Give two ways to communicate model uncertainty to a non-technical audience.

Q5. MCQ

A truncated y-axis that exaggerates a small difference is a problem of:

- (A) Visual integrity
- (B) Referential integrity
- (C) L2 regularisation
- (D) PCA scaling

Q6. Short Answer

Name two stakeholders who may be affected by a predictive policing model and one harm for each.

Q7. MCQ

Which documentation item best supports accountability?

- (A) Data source, assumptions, metric, limitations
- (B) Only final accuracy
- (C) Only model name
- (D) Only SQL syntax

Q8. Short Answer

Why can fairness metrics conflict with each other?

Q9. MCQ

Which final-exam topic most directly checks communication from patterns to action?

- (A) Interpreting association rules for policy
- (B) Counting Python files
- (C) Memorising variable names
- (D) Ignoring stakeholders

Q10. Long Question

A city plans to use accident data to recommend road-safety interventions. Discuss privacy, bias, validity of the mined patterns, and how to present recommendations responsibly.



Answer Key

W1 Intro Ds And Big Data

1. Semester 2 2023 Final (Q11, modified): Design thinking keeps the problem grounded in stakeholder needs, and it encourages iterative framing before choosing data or models. It helps avoid solving a technically neat but organisationally irrelevant problem.
2. Semester 2 2023 Final (Q12, modified): Examples: bad data can make a risk model under-serve a group with incomplete records; confusing correlation with causation can lead a policy team to fund an intervention that is only associated with, not causing, the desired outcome.
3. B. This is labelled response prediction.
4. Question, acquire, clean/transform, store, EDA, model/mine, evaluate, communicate. Acquisition risk: biased sample, invalid source, missing consent, or wrong row grain.
5. A. Suburb is nominal.
6. It may optimise the wrong objective, ignore stakeholder costs, be hard to explain, embed bias, or be unusable in workflow.
7. B.
8. Churn is per customer, so transaction rows need aggregation; otherwise frequent customers are over-weighted and labels may leak future behaviour.
9. B. Velocity.
10. Using future outcome-derived variables; scaling/imputing using the full dataset; including post-decision timestamps; duplicate customers across train/test.

W2 Spreadsheet Eda

1. Semester 1 2024 Final (Q2, modified): A. With an intercept, ordinary training R^2 is usually between 0 and 1; interpret it as explained variance proportion, not a causal claim.
2. Semester 2 2023 Final (Q14, modified): Total population is 80k. Proportions are A 1/8, B 3/8, C 4/8. Sample sizes are A 20, B 60, C 80.
3. B.
4. Mean = $32/5 = 6.4$, median = 3; median is better for typical value because 20 is an outlier.
5. A.
6. The numeric codes impose fake distances/order; models may treat blue-red as twice green-red etc.
7. $z = (65 - 50)/10 = 1.5$.
8. A.
9. For monotonic but nonlinear relationships or ordinal/ranked data, especially with outliers.
10. A.

W3 Python Eda

1. Semester 2 2023 Final (Q15, modified): Any two valid checks earn credit, e.g. incomplete data by column missing percentage; inconsistent data by unit/code validation; duplicates by primary/business key; out-of-date data by timestamp age; invalid data by domain/range rules.
2. A.
3. To avoid validation/test information influencing transformation parameters, which would inflate performance.
4. $140/2000 = 0.07 = 7\%$.
5. A.
6. Advantage: simple and transparent. Risk: biased estimates and lost power if missingness is not MCAR.
7. A.
8. It makes random splits/initialisation reproducible; it does not remove sampling bias or guarantee the best model.
9. C.
10. Check exact key duplicates and fuzzy duplicates; business rules decide whether same email/date or same id/version is duplicate.

W4 Clean And Store

1. Semester 2 2024 Final (Q1A, modified): Star schema keeps denormalised dimension tables around a central fact table, giving simpler/faster BI queries but redundancy. Snowflake normalises dimensions into sub-dimensions, reducing redundancy and



- improving integrity but requiring more joins.
2. A.
 3. Fact stores event measurements at a grain; dimensions describe who/what/where/when context.
 4. A.
 5. Snowflake reduces redundancy and can improve integrity; it needs more joins and is less simple for BI queries.
 6. 3 rows, because grain is product line per transaction.
 7. A.
 8. It is stable, compact, and avoids changes/collisions in natural identifiers.
 9. A.
 10. One row per student-lecture meeting, or per lecture aggregate; must be stated before measures.

W5 Sql Summary

1. Semester 1 2023 Final (Q2, modified): Join Tx to Tx_Item on transaction id, then join Tx_Item to Item on item id, filter Tx.month=3, and select DISTINCT item names.
2. Semester 2 2022 Final (Q7, modified): Join Station to Measurement on Station, group by station/site name, and select Site-Name with COUNT(m.Value) and AVG(m.Value).
3. B.
4. COUNT(*)=3; COUNT(A)=2.
5. B.
6. It joins on all same-named columns, including accidental ones, making results fragile and hard to audit.
7. One valid query joins Film to Film_Actor on film id, groups by film id and title, uses HAVING COUNT(actor_id)>5, and orders by title.
8. A.
9. WHERE filters rows before grouping, e.g. March sales; HAVING filters aggregated groups, e.g. stores with total sales above 1000.
10. A.

W6 Hyp Test And Eval

1. Semester 1 2024 Final (Q1, modified): A. Rejecting a true null hypothesis is a Type I error.
2. Semester 1 2023 Final (Q3, modified): Ask whether the mean favourability rating differs between flavours. $H_0 : \mu_A = \mu_B$. Randomly assign/taste both under a controlled design, collect numeric ratings such as 1–5, and use a paired or independent t-test depending on design; reject if $p < \alpha$.
3. A.
4. Reject H_0 because $0.005 < 0.01$; evidence against H_0 at 1 percent level.
5. $H_0 : \mu_1 = \mu_2$; $H_1 : \mu_1 \neq \mu_2$ for two-sided difference.
6. A.
7. Accuracy = $(30 + 40)/100 = 0.70$; precision = $30/(30 + 10) = 0.75$; recall = $30/(30 + 20) = 0.60$.
8. It estimates generalisation across folds; changing model complexity/features/regularisation/data quality is needed to fix fit problems.
9. A.
10. Independent groups, numeric response, approximate normality or adequate sample size, similar variances if using pooled t-test.

W7 Assoc And Pca

1. Semester 2 2022 Final (Q5, modified): Support threshold is 3 of 4 transactions. Frequent 2-itemsets include Apple, Banana in 4 transactions and Apple, Diapers, Banana, Diapers in 3 transactions. Other 2-itemsets have support below 3.
2. Semester 2 2024 Final (Q3, modified): The discarded components collectively explain at most 5 percent of variance. PCs are linear combinations of original variables, so removing 10 PCs is not the same as deleting 10 original features.
3. A.
4. Support = $30/100 = 0.30$; confidence = $30/40 = 0.75$.
5. If an itemset is frequent, all subsets are frequent; if a subset is infrequent, supersets cannot be frequent.
6. A.
7. If consequent is already common, confidence can be high without useful association; lift near 1 indicates little added value.



8. A.
9. Total variance = 10; first two explain $8/10 = 80\%$, so keep 2 PCs.
10. PCA is variance/distance based; large-scale variables dominate components if not standardised.

W8 Clustering

1. Semester 2 2024 Final (Q5, modified): First merge A and B at distance 2. Second merge C and D at distance 4, because it is the smallest remaining single-linkage distance. The next comparison would use minimum cross-distances between clusters.
2. Semester 2 2022 Final (Q3, modified): Choose initial prototypes/centroids, assign each point to the nearest prototype, update prototypes from assigned points, and iterate until stable. Choose/evaluate clusters using SSE elbow, silhouette, stability, or domain interpretability.
3. A.
4. 0,2 assigned to centroid 1; 10,12 to centroid 11. Updated centroids: 1 and 11.
5. Bad k, scaling problems, outliers, non-spherical clusters, local minima, random initialisation.
6. A.
7. Likely $k = 2$ or $k = 3$; largest drop is 1 to 2, smaller after 3, so justify with elbow plus silhouette/domain.
8. Near 1: close to own cluster and far from next. Near -1: likely assigned to wrong cluster.
9. A.
10. No labels or ground truth; results depend on distance/features/k and need interpretation/validation.

W9 Regression

1. Semester 2 2024 Final (Q3, modified): Simple regression has one predictor, e.g. $y = \beta_0 + \beta_1 x + \epsilon$. Multiple regression has several predictors, e.g. $y = \beta_0 + \sum_j \beta_j x_j + \epsilon$, and each coefficient is interpreted holding the other predictors fixed. Highly correlated predictors can cause multicollinearity and unstable coefficients.
2. Semester 2 2024 Final (Q3A, modified): False. Training R^2 usually cannot decrease when adding features, but the model may overfit or leak information; compare validation/test performance or adjusted R^2 .
3. A.
4. $SSE = (2 - 1)^2 + (4 - 5)^2 + (5 - 5)^2 = 2$.
5. Predictors share information; conditional effect can be unstable/near zero when other correlated predictors are included.
6. A.
7. Prediction = $1.115(4) - 1.077 = 3.383$; squared error $(3.1 - 3.383)^2 \approx 0.080$.
8. Simple coefficient is change per unit x; multiple coefficient is change per unit holding other predictors constant.
9. A.
10. It penalises unnecessary predictors, so it can decline when added features do not justify complexity.

W10 Classification

1. Semester 2 2022 Final (Q2, modified): Compute target entropy first. For each candidate attribute, compute the weighted conditional entropy after splitting by its values. Information gain is parent entropy minus weighted child entropy; choose the attribute with largest information gain as the root.
2. Semester 1 2024 Final (Q5, modified): False. Differentiability allows gradient updates, but non-convex losses may have local minima/saddles and convergence also depends on learning rate and optimisation conditions.
3. A.
4. $\sigma(0) = 0.5$; at threshold 0.5 classify positive if rule uses ≥ 0.5 .
5. Gradient descent needs gradients; nondifferentiable loss prevents ordinary smooth gradient updates.
6. A.
7. $p = 6/8 = 0.75$; entropy = $-0.75 \log_2 .75 - 0.25 \log_2 .25 \approx 0.81$.
8. IG is parent entropy minus weighted child entropy; choose split that most reduces impurity.
9. A.
10. $0.6 \times 0.5 \times 0.2 \times 0.8 = 0.048$.

W11 Text And Bayes

1. Semester 2 2022 Final (Q6, modified): Q vector is (1,1). D1 vector is (1,0). Manhattan distance is $|1 - 1| + |1 - 0| = 1$.
2. Semester 2 2024 Final (Q4, modified): Score = $0.5 \times 0.8 \times 0.4 \times 0.25 = 0.04$.



3. A.
4. It ignores order/context and semantics; sparse high-dimensional features; cannot capture negation unless n-gram.
5. $\log(100) = 4.605$ approximately.
6. A.
7. Headers may contain direct category/spam cues unavailable or artificial, causing leakage and poor generalisation.
8. A.
9. $0.4 \times 0.5 \times 0.3 = 0.06$.
10. False positives delete legitimate emails, often worse than letting some spam through.

W12 Ethics And Comms

1. A.
2. Collect only attributes necessary for the task; reduces exposure, misuse, and re-identification risk.
3. A.
4. Confidence intervals, error bars, prediction intervals, scenario ranges, plain-language caveats.
5. A.
6. Communities: over-policing; drivers/pedestrians: misallocated safety resources; city: unfair or ineffective policy.
7. A.
8. Because equalising one rate, e.g. FPR, can change another, e.g. FNR, when base rates differ.
9. A.
10. Assess consent/privacy, sample/reporting bias, confounding, support/lift stability; present recommendations as evidence-informed with uncertainty and monitoring plan.